

Random Hyperbolic Surface Spectral Rigidity — cycles 40-42

2026-05-16

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Random Hyperbolic Surface Spectral Rigidity — cycles 40-42

Abstract

Cycles 40-42 moved the campaign from direct consequences of Kim-Tao’s theorems into a scoped finite-model benchmark program. Cycle 40 completed one last Theorem 2 proof-mining step: it extracted a fixed-cutoff pre-Sobolev local L^2 mass estimate from the proof of eigenfunction delocalization. Cycles 41 and 42 then pivoted to the two-generator Schreier/random-permutation model as a controlled benchmark for trace-expansion mechanisms. Cycle 41 established a fixed- k trace-moment theorem template and reproducible variance evidence. Cycle 42 upgraded that variance evidence into a paired-word covariance mechanism, validating the next theorem target

$$\text{Var} \left(n^{-1} \text{Tr}(A_n^k) \right) = O_k(n^{-2})$$

for the Schreier benchmark, subject to a remaining all-fixed- k quotient-template lemma.

All three cycles were validated. M29 was accepted without repair. M30 and M31 were validated after moderate repairs to their analyzers and tests. No cycle claimed a new hyperbolic random-cover theorem beyond the scoped Kim-Tao consequences and Schreier benchmark statements.

Introduction

The broader campaign studies Kim and Tao’s paper 2603.01127, “Eigenvalue rigidity of hyperbolic surfaces in the random cover model.” Earlier cycles reconstructed the paper, built proof

ledgers and computational probes, and explored several extension routes. By the end of cycle 39, the direct theorem-consequence branches had produced:

- rigidity-scale multiplicity and cluster envelopes from Theorem 1;
- L^p , small-set mass, and effective-support corollaries from Theorem 2.

The cycle 39 audit recommended avoiding another direct interpolation or bookkeeping cycle unless the proof of Theorem 2 contained a genuinely sharper local-mass statement. Cycle 40 therefore mined the proof architecture of Theorem 2. After that route produced a scoped fixed-cutoff statement, cycles 41-42 shifted to a separate benchmark: random Schreier graphs generated by two independent random permutations. The purpose of this model is not to prove Kim-Tao surface results, but to provide a fully inspectable trace-expansion analogue where tree subtraction, fixed-point statistics, and variance mechanisms can be tested exactly or numerically.

Approach

The report follows the chronological arc of the three cycles.

Cycle 40, M29, inspected the proof of Theorem 2 upstream of the final L^∞ bound. The central question was whether the pre-trace formula and fourth-moment estimate already imply a standalone local L^2 mass estimate before the final Sobolev or elliptic conversion.

Cycle 41, M30, defined the Schreier graph model

$$A_n = P_a + P_a^{-1} + P_b + P_b^{-1},$$

where P_a and P_b are independent uniform permutations in S_n . It expanded traces into fixed points of words and compared normalized trace moments with the 4-regular tree moments.

Cycle 42, M31, expanded the M30 variance into paired fixed-word covariances. It used labelled constraint templates to test whether all fixed paired-word covariance terms remain $O_k(1)$ before the outer n^{-2} normalization.

The report cites the supplied session IDs for traceability:

Cycle	Role	Session ID	Main content
40	Researcher	fdd6b822-2434-482f-a181-d771d1ecf8ef	M29 brief for Theorem 2 pre-trace local mass
40	Worker	9747c28e-f40b-48d7-907c-0d6b76526f81	M29 artifacts, scripts, data, figures
40	Auditor	1292619c-c231-4b36-89f7-4b51440cf9b4	M29 validation
41	Researcher	1c2a42a8-f835-4d19-87a0-d6a88b9ee390	M30 Schreier benchmark brief
41	Worker	b4b2c51f-475d-4120-bae7-d8c7ea5f4d3e	M30 artifacts, scripts, data, figures
41	Auditor	02853112-349d-48a1-8789-92f7635f8ce7	M30 validation after repair
42	Researcher	e41a60af-48f1-4a6c-8cec-b00c87acd419	M31 variance-mechanism brief

Cycle	Role	Session ID	Main content
42	Worker	36bd1898-365c-464f-a154-af593ceb90e6	M31 artifacts, scripts, data, figures
42	Auditor	2eafdd6e-57f2-4e58-ae05-99b76864ce28	M31 validation after repair

Source Inventory and Timeline

Cycle 40 began with the M29 researcher brief. It asked whether Kim-Tao’s Theorem 2 proof contains a local mass estimate before the final sup-norm theorem. The required chain was:

pre-trace formula $\rightarrow$ centered local spectral mass $\rightarrow V_n \rightarrow S$ -subtraction $\rightarrow$ Proposition 4.2 $\rightarrow$ Proposition

The worker produced docs/proof_ledger/pretrace_local_mass_intermediate.md, reports/extension_candidates/m29_pretrace_local_mass_intermediate.md, a final follow-up statement, one analyzer, one test file, two CSV files, and two figures. The auditor validated the result without repair, citing 2603.01127.txt lines 1167-1229 as the source region supporting Proposition 4.1, Chebyshev/fiber-window union, and the later Sobolev/elliptic loss.

Cycle 41 then opened M30. The researcher brief moved from Kim-Tao theorem consequences to a finite Schreier benchmark. The worker built docs/proof_ledger/schreier_trace_moment_benchmark.md, reports/extension_candidates/m30_schreier_benchmark_theoremization.md, a trace benchmark analyzer, a test file, four CSV files, and three figures. The auditor found one moderate defect: the analyzer’s classification logic used hard-coded default n values, so custom grids could downgrade the branch decision incorrectly. The repair made the classifier derive the terminal n from the current variance table and added regression coverage. After repair, M30 was validated.

Cycle 42 opened M31 as a variance theoremization step. The worker built docs/proof_ledger/schreier_variance_mechanism.md, reports/extension_candidates/m31_schreier_variance_mechanism.md, a pair-template analyzer, a test file, four CSV files, and three figures. The auditor found one moderate defect: the analyzer claimed class-level $0(1)$ covariance support using a hardcoded nonidentity bound while detailed exponents were recorded for only one representative per class. The repair made the analyzer compute maximum distinct-basepoint and same-basepoint template exponents across all reduced word-pair templates in each checked class. After regeneration and added tests, M31 was validated.

No REFERENCES.md file exists in the workspace. The References section therefore lists local paper files, session records, reports, proof notes, data artifacts, figures, and audit inputs rather than global numbered bibliography entries.

Finding 1: M29 Extracted a Fixed-Cutoff Local Mass Corollary from Theorem 2

M29 found that Theorem 2 contains a proof-level local L^2 mass estimate before the final L^∞ conversion. The extracted statement is a fixed-cutoff result, not a uniform geometric ball theorem.

The controlled object is the centered pre-trace local spectral mass, raised to a fourth-power statistic V_n , with a primitive-power diagonal term S subtracted before Proposition 4.1 and

Proposition 4.2 are applied. On the Theorem 2 high-probability event, for the fixed nonnegative cutoff a in $C_c^\infty(H)$ used in the proof, every fiber i and every normalized eigenfunction below energy Λ_0 satisfies the schematic estimate

$$\int_{\mathbb{H}} a(z) |u_j^\rho(z, i)|^2 d\text{Vol}(z) \leq C\Lambda_0 n^{-\alpha_0}, \quad \alpha_0 = \frac{1}{16(2\kappa + 11)}.$$

The M29 classification table records this as `standalone_smoothed_kernel_mass_corollary` with branch decision `advance_pretrace_local_mass_branch`.

M29 compared this statement with the M28 deterministic mass bound obtained from the final sup-norm theorem. M28's final bound gives, for a fixed-volume patch in schematic form,

$$\int_A |u|^2 \leq C\Lambda_0^3 n^{-\alpha_0} \text{vol}(A).$$

The pre-Sobolev estimate has Λ_0 power 1 rather than 3. At fixed energy this is a theorem-strength distinction in the energy dependence rather than an improved n exponent. If $\Lambda_0 = n^b$, the improvement is n^{2b} in the bookkeeping model.

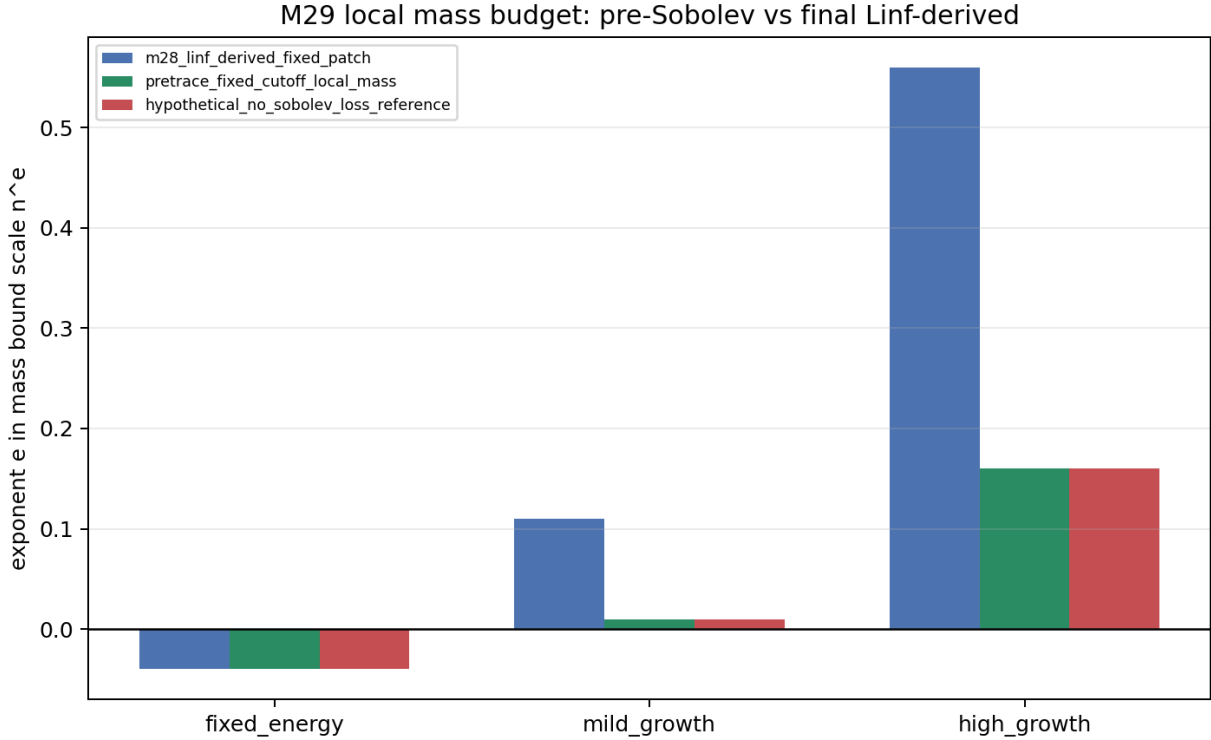


Figure 1: M29 comparison of M28 final sup-norm-derived mass bounds, the extracted pre-trace fixed-cutoff estimate, and a comparison-only no-Sobolev-loss reference.

The scope boundary is central. The proof fixes a smooth cutoff a ; it does not union over all spatial centers, dominate arbitrary nonsmooth ball indicators, or imply lower mass. The auditor explicitly validated that M29 does not claim quantum ergodicity, random-wave behavior, lower mass on balls, shrinking-ball control, nodal information, or equidistribution.

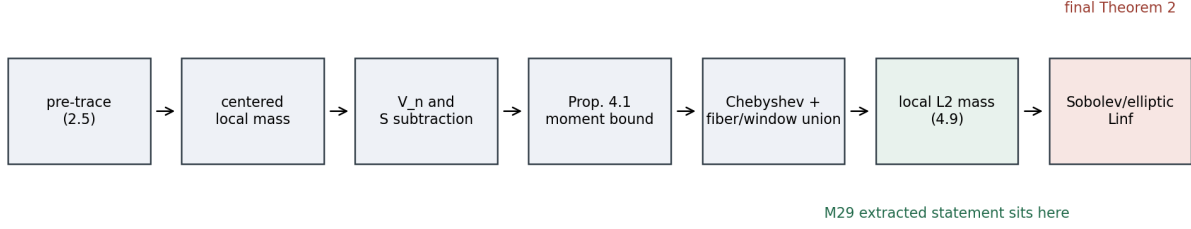


Figure 2: M29 proof-pipeline figure locating the fixed-cutoff local mass estimate before the final Sobolev/elliptic conversion.

Finding 2: M30 Turned the Schreier Probe into a Trace-Moment Benchmark

M30 shifted the campaign away from direct Kim-Tao theorem corollaries and into a finite Schreier/random-permutation benchmark. The model uses the adjacency operator

$$A_n = P_a + P_a^{-1} + P_b + P_b^{-1}$$

on n vertices. For fixed k , the trace expands as

$$\text{Tr}(A_n^k) = \sum_{w \in \{a, a^{-1}, b, b^{-1}\}^k} \text{Fix}(w(P_a, P_b)).$$

Freely reducible words act as the identity and supply the 4-regular tree moment. Nontrivial reduced words have $0_k(1)$ expected fixed points, so the normalized expectation has the theorem-template form

$$\mathbb{E}[n^{-1} \text{Tr}(A_n^k)] = m_k + O_k(n^{-1}).$$

The tree moments were regenerated exactly through $k=10$:

k	Tree moment
0	1
2	4
4	28
6	232
8	2092
10	19864

Odd tree moments vanish in the generated table. Dynamic programming and free-reduction enumeration agreed.

M30 also produced reproducible simulation evidence using $n=80, 140, 220, 320$, 24 trials, and seed 20260516, for 960 trial rows. The centered variance slopes for the main even moments were:

k	Fitted variance slope
2	-1.765
4	-1.716
6	-1.762

M30 classified the fixed-k expectation as a theorem template and the variance scaling as numerical evidence. The branch decision was `advance_schreier_benchmark_program`.

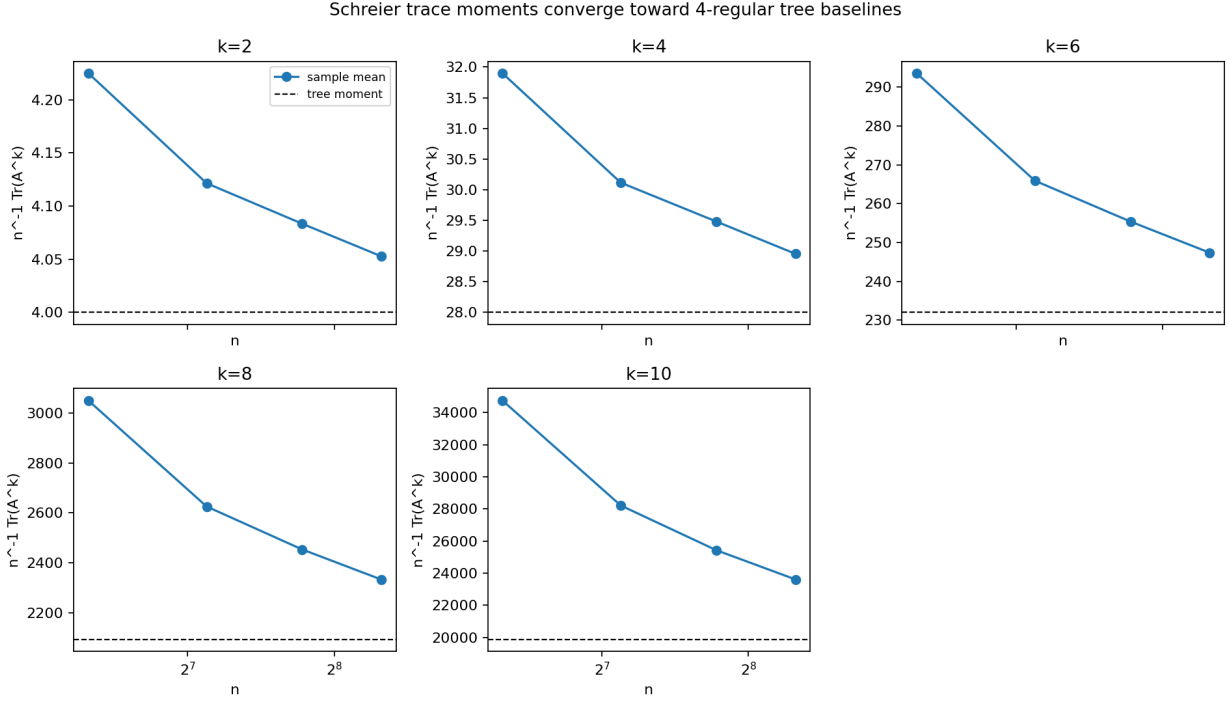


Figure 3: M30 normalized Schreier trace moments versus n , with tree moment baselines.

The M30 audit repaired only the custom-grid classifier. It did not change the default conclusion. The validated result is a scoped benchmark, not evidence for a hyperbolic random-cover transfer. The model has a free-group random permutation law and adjacency spectrum, not a surface-group cover law or hyperbolic Laplacian.

Finding 3: M31 Validated the Schreier Variance Mechanism as the Next Theorem Target

M31 upgraded M30's empirical variance evidence into a paired-word covariance mechanism. The starting identity is

$$\text{Var} \left(n^{-1} \sum_w \text{Fix}(w) \right) = n^{-2} \sum_{w_1, w_2} \text{Cov}(\text{Fix}(w_1), \text{Fix}(w_2)).$$

The key theorem template is: if every fixed nontrivial paired-word covariance is $0_k(1)$, then

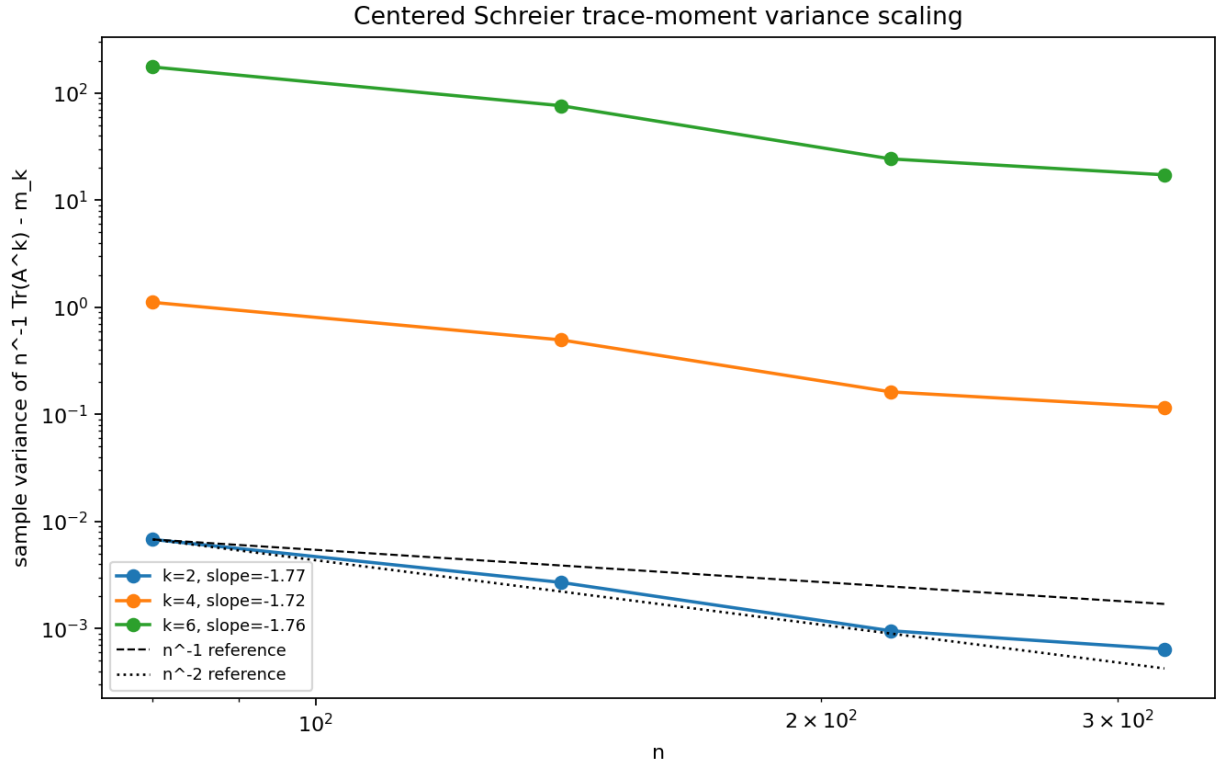


Figure 4: M30 centered trace-moment variance scaling with fitted slopes and reference n^{-1} and n^{-2} slopes.

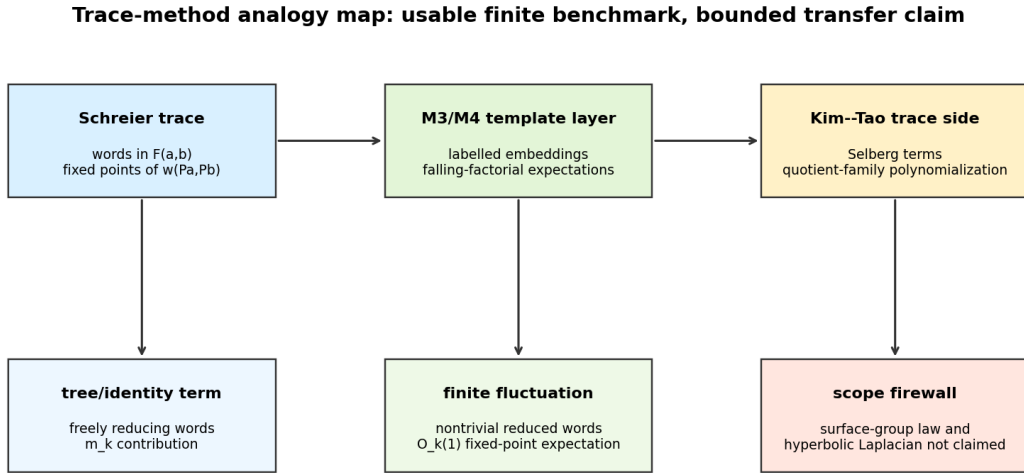


Figure 5: M30 analogy map comparing the Schreier trace expansion, earlier labelled-template identities, and Kim-Tao trace-side polynomialization.

$$\text{Var}(n^{-1} \text{Tr}(A_n^k)) = O_k(n^{-2}).$$

M31 checked reduced word-pair templates for $k=2,4,6$. After audit repair, class-level order claims were computed from all reduced pair templates in each checked class, not from one representative. The repaired summary is:

k	Word pairs	Reduced word types	Pair classes	Max covariance order before normalization	Normalized order
2	256	13	7	$0(1)$	$0(n^{-2})$
4	65,536	121	8	$0(1)$	$0(n^{-2})$
6	16,777,216	1,093	8	$0(1)$	$0(n^{-2})$

No checked class had a positive-power covariance obstruction. Identity-containing classes were treated separately as deterministic covariance-zero terms. The detected relation classes include same reduced words, inverse reduced words, cyclic or inverse-conjugate pairs, shared powers, and generic pairs.

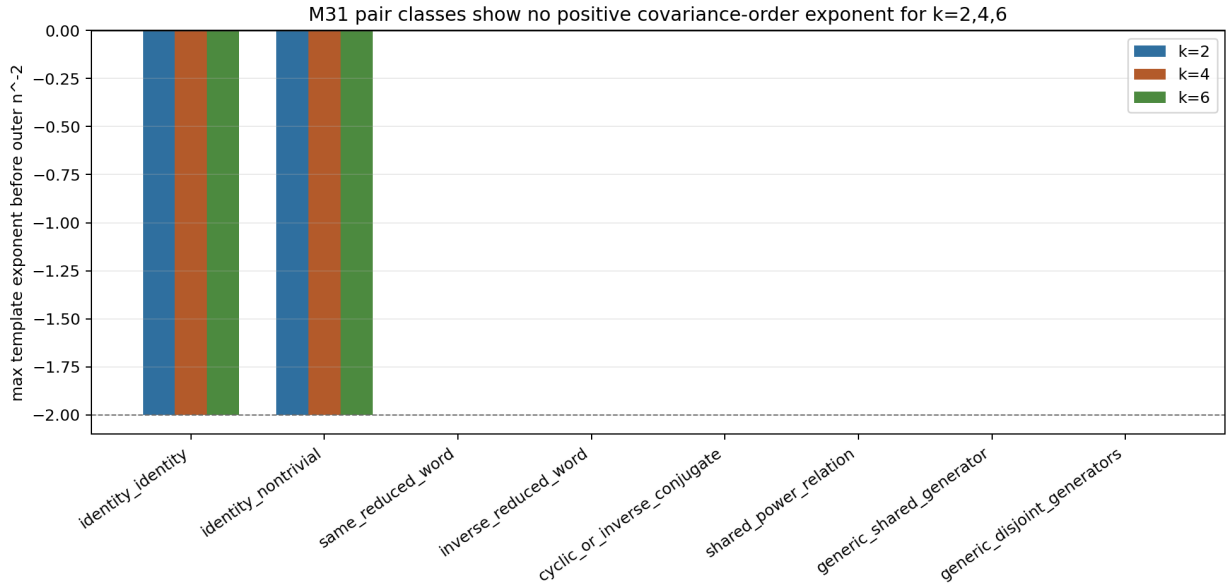


Figure 6: M31 pair classes versus covariance-order exponents for $k=2,4,6$.

M31 also reinterpreted the M30 slopes near -1.7 . Since the pair-template evidence found no $0(n)$ covariance mechanism in checked classes, the validated interpretation is finite-size crossover toward an n^{-2} asymptotic, not evidence for a genuine n^{-1} covariance law.

The remaining theorem gap is explicit:

For every fixed nontrivial reduced pair (u, v) , every consistent quotient template contributing to $\mathbb{E} \text{Fix}(u)$

Schreier variance mechanism: finite word-pair covariance reduces to labelled templates

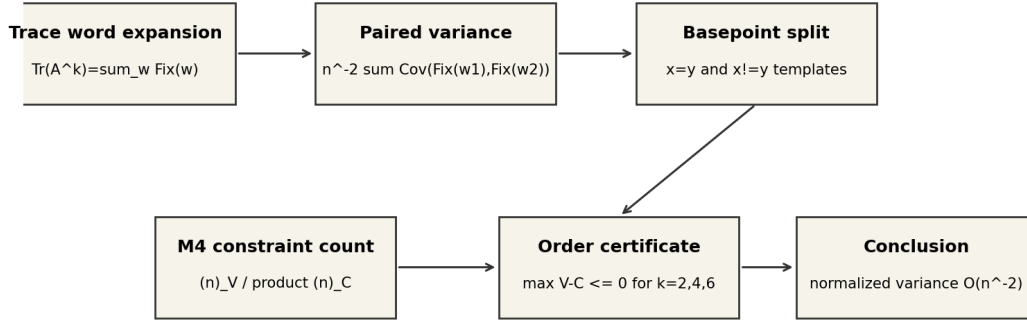


Figure 7: M31 mechanism map from paired trace expansion to labelled-template order certification.

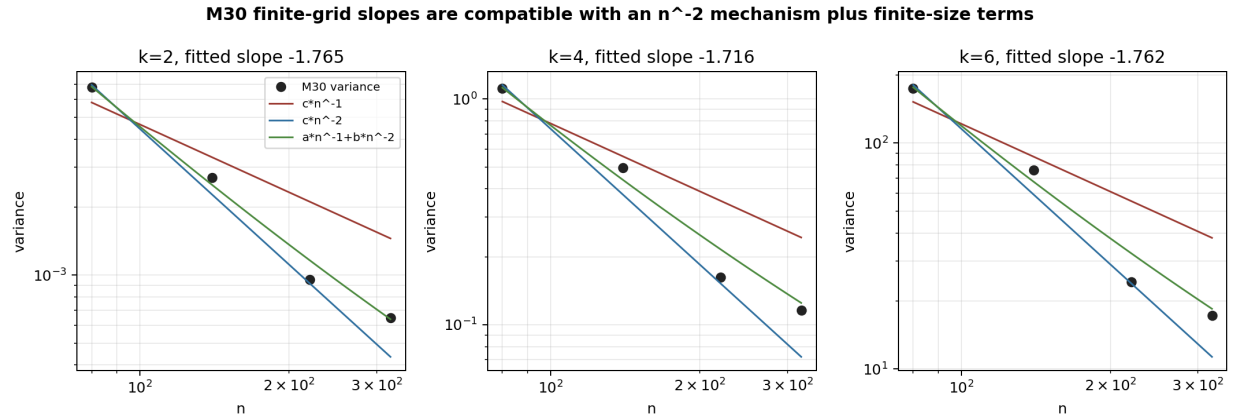


Figure 8: M31 reinterpretation of M30 fitted slopes against n^{-1} , n^{-2} , and mixed finite-size models.

A proof of this finite quotient-template exponent lemma would turn the M31 theorem template into a fixed- k Schreier variance theorem. M31 does not transfer this conclusion to Kim-Tao random covers; doing so would require a new surface-group quotient-family theorem.

Discussion

The three-cycle sequence has two parts.

First, M29 closes the last direct Theorem 2 proof-consequence branch. It shows that the proof contains a sharper fixed-cutoff local mass statement than the final sup-norm corollary alone reveals. This is a genuine theorem-level extraction, but it remains tied to the proof's chosen smooth cutoff and does not become a spatial equidistribution theorem.

Second, M30-M31 establish the Schreier benchmark as the current productive branch. M30 supplies the fixed- k trace-moment baseline and reproducible variance evidence. M31 explains that evidence through paired fixed-word covariance templates and identifies the exact finite combinatorial lemma still needed for a theorem.

The analogy with Kim-Tao is structural. In both settings, one expands a trace or two-trace statistic, separates deterministic or diagonal contributions, and reduces random fluctuation control to fixed-point or quotient-template counting. The analogy stops before any hyperbolic conclusion: the Schreier model has independent free generators, while Kim-Tao's random covers involve surface-group relations, Selberg or pre-trace formula inputs, and external quotient-family machinery.

Open Questions

The immediate next question is the M31 fixed-pair exponent lemma. Proving it for arbitrary fixed nontrivial reduced word pairs would complete the finite Schreier variance theorem target.

A second route is to use M30-M31 as the validated benchmark base for finite non-shrinking spectral statistics. That route would remain finite-model work unless a separate surface-group bridge is developed.

The M29 local-mass result also leaves a proof-theoretic question: whether the fixed smooth cutoff can be replaced by a controlled finite family of geometric patches. The current cycles do not establish that extension.

The campaign should still avoid the unsupported claims excluded by the audits: quantum ergodicity, random-wave behavior, lower mass on arbitrary balls, shrinking-ball statistics, nodal information, simplicity, level repulsion, or hyperbolic local statistics from M27-M31 alone.

References

No REFERENCES.md file was present in the workspace at report time. The sources used for this report are therefore local campaign sources:

- Kim-Tao source files: 2603.01127.pdf, 2603.01127.txt.
- Cycle 40 sessions: researcher fdd6b822-2434-482f-a181-d771d1ecf8ef, worker 9747c28e-f40b-48d7-907c-0d6b76526f81, auditor 1292619c-c231-4b36-89f7-4b51440cf9b4.
- Cycle 41 sessions: researcher 1c2a42a8-f835-4d19-87a0-d6a88b9ee390, worker b4b2c51f-475d-4120-bae7-d8c7ea5f4d3e, auditor 02853112-349d-48a1-8789-92f7635f8ce7.
- Cycle 42 sessions: researcher e41a60af-48f1-4a6c-8cec-b00c87acd419, worker 36bd1898-365c-464f-a154-af593ceb90e6, auditor 2eafdd6e-57f2-4e58-ae05-99b76864ce28.

- Proof ledgers: docs/proof_ledger/pretrace_local_mass_intermediate.md, docs/proof_ledger/schreier_trace_moment_benchmark.md, docs/proof_ledger/schreier_variance_mechanism.md.
- Extension reports: reports/extension_candidates/m29_pretrace_local_mass_intermediate.md, reports/extension_candidates/m30_schreier_benchmark_theoremization.md, reports/extension_candidates/m31_schreier_variance_mechanism.md.
- Data artifacts under data/extension_candidates/ for M29-M31.
- Figures under reports/figures/ for M29-M31.
- Supplied cycle 42 audit report.

Appendix: Implementation Details

Code Organization

Cycle 40 added the M29 pre-trace local-mass package:

File	Lines	Purpose
scripts/analyze_pretrace_local_mass_budget.py	224	Compares M28 final sup-norm mass bounds, M29 pre-trace local-mass bounds, and comparison-only models
tests/test_pretrace_local_mass_budget.py	76	Checks classification uniqueness, source-backed claims, comparison-only safeguards, and unsupported-label exclusion
docs/proof_ledger/pretrace_local_mass_intermediate.md	85	Records Theorem 2 proof pipeline and extracted statistic
reports/extension_candidates/m29_pretrace_local_mass_intermediate.md	51	States M29 branch decision and scope
reports/final/pretrace_local_mass_followup_statement.md	17	Concise follow-up statement

Cycle 41 added the M30 Schreier trace benchmark package:

File	Lines	Purpose
scripts/analyze_schreier_trace_benchmark.py	417	Regenerates tree moments, simulations, variance slopes, classifications, and figures
tests/test_schreier_trace_benchmark.py	87	Checks tree moments, seed reproducibility, classification, and custom-grid repair

File	Lines	Purpose
docs/proof_ledger/schreier_trace_moment_benchmark.md	89	Defines the Schreier model and fixed-k trace-moment theorem template
reports/extension_candidates/m30_schreier_benchmark_theoremization.md	87	Packages the benchmark and analogy firewall
reports/final/schreier_benchmark_followup_statement.md	23	Concise follow-up statement

Cycle 42 added the M31 Schreier variance mechanism package:

File	Lines	Purpose
scripts/analyze_schreier_variance_pair_templates.py	597	Enumerates pair classes and classwise covariance template exponents
tests/test_schreier_variance_pair_templates.py	95	Checks free reduction, identity handling, relation classes, and all-pair class-order computation
docs/proof_ledger/schreier_variance_mechanism.md	112	Records paired-word variance expansion and theorem target
reports/extension_candidates/m31_schreier_variance_mechanism.md	75	Packages M31 result and scope firewall
reports/final/schreier_variance_followup_statement.md	21	Concise follow-up statement

Generated Data

M29 generated:

File	Lines	Contents
data/extension_candidates/m29_pretrace_local_mass_budget.csv	19	18 budget rows plus header
data/extension_candidates/m29_local_mass_statement_classification.csv	7	Six classification rows plus header

M30 generated:

File	Lines	Contents
data/extension_ candidates/m30_schreier_ tree_moments.csv	12	Tree moments through k=10 plus header
data/extension_ candidates/m30_schreier_ trace_moment_trials.csv	961	960 simulation rows plus header
data/extension_ candidates/m30_schreier_ variance_scaling.csv	41	40 variance rows plus header
data/extension_ candidates/m30_schreier_ benchmark_ classification.csv	6	Five classification rows plus header

M31 generated:

File	Lines	Contents
data/extension_ candidates/m31_pair_ template_classes.csv	24	Pair-class table plus header
data/extension_ candidates/m31_pair_ covariance_orders.csv	24	Covariance-order table plus header
data/extension_ candidates/m31_variance_ order_summary.csv	4	k=2,4,6 summary rows plus header
data/extension_ candidates/m31_variance_ mechanism_ classification.csv	6	Five classification rows plus header

Figure Inventory

The report uses all cycle 40-42 generated figures inline:

Figure	Size
reports/figures/m29_local_mass_exponent_ comparison.png	1440 x 900
reports/figures/m29_theorem2_proof_ pipeline.png	1980 x 576
reports/figures/m30_schreier_moment_ convergence.png	1920 x 1120
reports/figures/m30_schreier_variance_ scaling.png	1360 x 880
reports/figures/m30_trace_method_analogy_ map.png	1760 x 928

Figure	Size
reports/figures/m31_pair_class_covariance_orders.png	1980 x 972
reports/figures/m31_variance_order_mechanism_map.png	1800 x 864
reports/figures/m31_m30_slope_reinterpretation.png	2160 x 756

Validation Results

Final post-manifest validation was run after this reporter pass updated MANIFEST.md.

`python3 -m long_exposure.tools.promise_check .` passed with:

```
events: 121, plan milestones: 31
```

Warnings were historical: one noncanonical `docs/paper_map/ ledger` path and orphan prior periodic reports under `reports/cycles/`.

`python3 -m long_exposure.tools.org_check .` passed with historical warnings for root paper/live-run files and old figures under `docs/`.

MANIFEST.md was replaced with a current cycles 40-42 snapshot. It now records:

- 40 Python scripts, 11,540 script lines;
- 32 Python tests, 2,634 test lines;
- 84 CSV datasets;
- 75 PNG figures under `reports/figures/`;
- 177 Markdown/DOT/PNG documentation/report artifacts;
- 121 promise ledger events;
- 31 plan milestones.

Cross-Reference Map

M28 final sup-norm mass corollaries feed into M29's proof-mining comparison. M29 extracts the fixed-cutoff pre-Sobolev local mass estimate and then closes the direct Theorem 2 consequence branch.

M3 and M4 supply the earlier Schreier computational and labelled-template context. M30 converts that context into a fixed-k trace-moment benchmark. M31 uses the M30 trace expansion to derive the paired-word variance mechanism.

The validated next step is either to prove M31's all-fixed-k quotient-template exponent lemma or to use M30-M31 as the benchmark base for finite non-shrinking spectral statistics. Neither path currently transfers to Kim-Tao hyperbolic random covers without a new surface-group quotient-family theorem.